# The Set of Sets Is Not the Same: Divisor Lattice Coherence and a Structural Reduction of Möbius Cancellation

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**Abstract**

Starting from the principle that *a collection of individually symmetric objects does not inherit their symmetry* — "the set of sets is not the same" — we construct a symmetry defect matrix $S_N(s)$ from the divisor lattices of the positive integers and investigate its spectral connection to the Riemann zeta function.

**Rigorous results.** We prove: (A) $S_N = D_s\, T_N\, D_s$ with $T_N[m,n] = f_s(\gcd(m,n))$ (factorization); (B) $\mathrm{rank}(S_N(s)) = Q(N) \sim 6N/\pi^2$ independently of $s$ (rank theorem); (C) $\mathrm{trace}(S_N(s)) \to \zeta(2s)/\zeta(-s)$ for $\mathrm{Re}(s) > 1/2$ (trace formula, via Euler product and analytic continuation); (D) $\|S_N(\sigma)\|_F^2 = \Theta(N^{2-4\sigma})$ (Frobenius norm); (E) $|\mathrm{trace}(S_N(\sigma))| = O(\sqrt{N})$ for real $\sigma > 1/2$ (rank–Frobenius bound); (F) $S_N(1-\sigma) = D^{-1}(S_N(\sigma) \circ R_N)D^{-1}$ exactly, with $R_N$ positive semi-definite (Hadamard identity); (G) $T_N = \bigcirc_p T_N^{(p)}$ where each $T_N^{(p)}$ has rank 2 (local Hadamard factorization — the matrix Euler product); (H) $\mathrm{rank}(T_N^{(d)}) = Q(N)$ for all $d \geq 2$ (dimensional invariance); (I) the truncated cancellation ratio $R_{\mathcal{P}}(\alpha)$ is strictly decreasing in $\alpha$ (Euler product monotonicity); (J) the per-prime cross term $\mathrm{CROSS}_p < 0$ for all primes $p$ (local variance reduction via Cauchy–Schwarz).

**Formal / heuristic results.** Via a formal Perron decomposition, the trace splits into a pole term $N^{1-2\sigma}$ and oscillatory terms $\sum c_\rho N^{\sigma+\rho}$. A zero off the critical line would produce anomalously large growth $\Theta(N^{1-\mathrm{Re}(\rho)})$.

**Numerical observations.** $\sigma_{\min}(S_N(1/4+it))$ collapses at $t \approx \gamma_k/2$, detecting the first zeros of $\zeta$. Eigenvalue cancellation exceeds 99.7% at $N = 5000$. Primes alone are insufficient for Nyman–Beurling ($d \approx 0.228$). The deformed trace $T(N,\alpha)$ satisfies $|T(N,1)| < |T(N,\alpha)|$ for all $\alpha \in (0,1)$ and $N \leq 100{,}000$.

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# 1 Introduction

## 1.1 The Riemann Hypothesis

The Riemann Hypothesis (RH), proposed in 1859 [6], asserts that all nontrivial zeros of the Riemann zeta function

$$\zeta(s) = \sum_{n=1}^{\infty} n^{-s}, \qquad \mathrm{Re}(s) > 1,$$

continued meromorphically to $\mathbb{C} \setminus \{1\}$, satisfy $\mathrm{Re}(s) = 1/2$. After more than 160 years and over $10^{13}$ numerically verified zeros, the conjecture remains unproven.

**Notation.** Throughout, $\mathbb{N}^* = \{1, 2, 3, \ldots\}$ denotes the positive integers, $P$ the set of primes, $\omega(n)$ the number of distinct prime factors of $n$, $\mu$ the Möbius function ($\mu(n) = (-1)^{\omega(n)}$ for squarefree $n$, $\mu(n) = 0$ otherwise), and $Q(N) = |\{k \leq N : \mu(k) \neq 0\}|$. We write $s = \sigma + it$ with $\sigma = \mathrm{Re}(s)$. The Hadamard (entrywise) product of matrices is denoted $\circ$.

## 1.2 The guiding principle: the set of sets is not the same

Our approach originates from a principle in the foundations of set theory:

*Principle* 1.1 (The set of sets is not the same). An individually symmetric collection of objects does not inherit a global symmetry. The failure of inheritance carries structural information.

This principle manifests at three levels in mathematics:

- **Cantor:** $|\mathcal{P}(A)| > |A|$ — the power set transcends the set.
- **Russell:** The collection of all sets is not itself a set.
- **Gödel:** A formal system cannot prove its own consistency.

We apply this principle to the divisor lattices of the positive integers. Each lattice $D(n) = \{d \in \mathbb{N}^* : d \mid n\}$ admits a natural involution $\tau_n : d \mapsto n/d$, but the family $\{D(n)\}_{n \in \mathbb{N}^*}$ does not inherit a single global involution. We quantify this *symmetry defect* through a matrix $S(s)$ and show that its spectral properties encode the zeros of $\zeta$.

## 1.3 Set-theoretic hierarchy

Our construction traverses five levels of set-theoretic objects:

| Level | Object | Type | Symmetry |
|---|---|---|---|
| 0 | $P$ (primes) | subset of $\mathbb{N}^*$ | none |
| 1 | $\mathbb{N}^*$ (integers) | multiplicative monoid | generated by $P$ |
| 2 | $\{D(n)\}_{n \in \mathbb{N}^*}$ | family of lattices | each $\tau_n$ is an involution |
| 3 | $S(s)$ (defect matrix) | $\mathbb{N}^* \times \mathbb{N}^*$ matrix | trace $= \zeta(2s)/\zeta(-s)$ |
| 4 | $Z$ (zeros of $\zeta$) | discrete subset of $\mathbb{C}$ | RH $\Leftrightarrow Z \subset \{\mathrm{Re} = 1/2\}$ |

Each arrow between levels *changes the nature of the object* — this is the content of Principle 1.1. The passage from Level 2 to Level 3 is where the zeros of $\zeta$ emerge: they measure the *excess* that the collection carries beyond its individual members.

## 1.4 Summary of results

**Rigorous theorems:**

**(A)** **Factorization** (Theorem 3.1): $S_N = D_s\, T_N\, D_s$ with GCD matrix $T_N$.

**(B)** **Trace formula** (Theorem 3.2): $\mathrm{trace}(S_N(s)) \to \zeta(2s)/\zeta(-s)$ for $\mathrm{Re}(s) > 1/2$.

**(C)** **Rank independence** (Theorem 3.4): $\mathrm{rank}(S_N(s)) = Q(N) \sim 6N/\pi^2$ for all $s$.

**(D)** **Frobenius norm** (Theorem 4.2): $\|S_N(\sigma)\|_F^2 = \Theta(N^{2-4\sigma})$ unconditionally.

**(E)** **Rank–Frobenius bound** (Proposition 5.1): $|\mathrm{trace}(S_N(\sigma))| = O(\sqrt{N})$ for real $\sigma > 1/2$.

**(F)** **Ramanujan decomposition** (Theorem 11.1): $(f_s * \mu)(d) = \mu(d)\, d^{3s}$.

**(G)** **Band concentration** (Theorem 11.3): hyperbolic bands $j > \sqrt{N}$ contribute $O(N^{7/8})$ to the error, reducing RH to $\sqrt{N}$ bands.

**(H)** **Hadamard identity** (Theorem 11.5): $S_N(1-\sigma) = D^{-1}(S_N(\sigma) \circ R_N)D^{-1}$, with $R_N$ PSD.

**(I)** **Local factorization** (Theorem 11.9): $T_N = \bigcirc_p T_N^{(p)}$, each of rank 2.

**(J)** **Dimensional invariance** (Theorem 11.25): $\mathrm{rank}(T_N^{(d)}) = Q(N)$ for all $d \geq 2$.

**(K)** **Euler product monotonicity** (Theorem 11.13): $R_{\mathcal{P}}(\alpha)$ is strictly decreasing in $\alpha$.

**(L)** **Finite cancellation derivative** (Theorem 11.14): $d\rho/d\alpha = -(2/\alpha)\, P_E\, P_O\, (\mu_E - \mu_O)$; monotonicity of $\mathcal{C}$ reduces to $\mu_E > \mu_O$ (verified for $N \leq 100{,}000$).

**(M)** **Variance decomposition** (Proposition 11.17): $(\sum f(k)w_k)^2 = \|w\|^2 + 2\,\mathrm{CROSS}$; for $f = \mu$, the cross term is negative (70%–97% reduction, numerical observation).

**(N)** **Local negativity per prime** (Theorem 11.20): $\mathrm{CROSS}_p < 0$ for every prime $p$ and every $N \geq 2p$; the global variance reduction decomposes into per-prime contributions, each provably negative by Cauchy–Schwarz.

**(O)** **W–CROSS identity** (Proposition 11.23): the monotone functional $W(a) = E^2/(E^2 + a\|w\|^2)$ and the normalized cross term satisfy $1 + 2\,\mathrm{CROSS}/\|w\|^2 = aW/(1-W)$ exactly.

**Formal / heuristic results** (based on unverified Perron bounds):

**(a)** **Singularity at zeros** (Proposition 3.6, formal): trace grows as $N^{1-\mathrm{Re}(\rho)}$ at $s = \rho/2$.

**(b)** **Anomalous growth detection** (Proposition 4.1, formal): a zero off the critical line produces detectable growth.

**Numerical observations:**

**(i)** Zero detection via $\sigma_{\min}$ collapses (Section 8).

**(ii)** Nyman–Beurling: primes alone stagnate at $d \approx 0.228$ (Section 6).

**(iii)** Eigenvalue cancellation $> 99.7\%$ at $N = 5000$ (Section 11.3).

**(iv)** $|T(N,1)| < |T(N,\alpha)|$ for all $\alpha \in (0,1)$, $N \leq 100{,}000$.

**(v)** Operator-norm transfer (Remark 11.7): open.

**(vi)** Log-concavity of $T_r$ and $\mu_E > \mu_O$ for $N \leq 100{,}000$ (Section 11.5).

**(vii)** $W(a)$ is strictly decreasing in $a$ for $N \leq 100{,}000$ with zero violations (Section 11.8).

# 2 Construction from set theory

All objects are constructed within ZFC (Zermelo–Fraenkel with Choice).

**Definition 2.1** (Divisor lattice and Euler involution)**.** For $n \in \mathbb{N}^*$, the *divisor lattice* is $D(n) = \{d \in \mathbb{N}^* : d \mid n\}$, partially ordered by divisibility. The *Euler involution* is $\tau_n : D(n) \to D(n)$, $d \mapsto n/d$. It satisfies $\tau_n^2 = \mathrm{id}$ and reverses the partial order.

**Definition 2.2** (Valuation space)**.** The *valuation map* $v : \mathbb{N}^* \to \mathbb{N}^{(P)}$ sends $n = \prod p_i^{a_i}$ to $(a_1, a_2, \ldots)$. This is an isomorphism of monoids: $(\mathbb{N}^*, \times) \cong (\mathbb{N}^{(P)}, +)$, where $\mathbb{N}^{(P)}$ denotes finitely supported functions $P \to \mathbb{N}$.

The multiplicative structure of $\mathbb{N}^*$ is encoded in the *direct sum* $V = \bigoplus_{p \in P} \mathbb{N}_p$, where $\mathbb{N}_p$ denotes a copy of $\mathbb{N}$ indexed by the prime $p$. The valuation map $v : \mathbb{N}^* \to V$ sends $n = \prod p_i^{a_i}$ to the tuple $(v_p(n))_{p \in P}$, where $v_p(n)$ is the $p$-adic valuation of $n$ (i.e., the exponent of $p$ in the factorization of $n$). The Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$ reflects this decomposition.

**Definition 2.3** (Symmetry defect matrix)**.** For $s \in \mathbb{C}$ and $N \in \mathbb{N}^*$, define $S_N(s) \in \mathbb{C}^{N \times N}$ by

$$S_N(s)[m,n] \;=\; \sum_{d \mid \gcd(m,n)} \mu(d)\, d^{\,s}\, (mn/d^2)^{-s}. \tag{1}$$

*Remark* 2.4 (Role of $\mu$)*.* The Möbius function $\mu(n) = (-1)^{\omega(n)}$ for squarefree $n$ (zero otherwise) is the *inclusion–exclusion kernel* on divisor sets. It selects the *irreducible* contribution of each common divisor $d$ to the interaction between $D(m)$ and $D(n)$. Combinatorially, $\mu$ is the *signed characteristic function* of squarefree integers — the set-theoretically "fundamental" elements of $\mathbb{N}^*$.

# 3 Structural theorems

## 3.1 Factorization

**Theorem 3.1** (Factorization)**.** *$S_N(s) = D_s\, T_N(s)\, D_s$ where $D_s = \mathrm{diag}(m^{-s})_{m=1}^N$ and $T_N(s)$ is the GCD matrix*

$$T_N(s)[m,n] = f_s(\gcd(m,n)), \qquad f_s(g) = \prod_{p \mid g} (1 - p^{3s}). \tag{2}$$

*Proof.* $S_N(s)[m,n] = m^{-s}\big[\sum_{d \mid \gcd(m,n)} \mu(d)\, d^{3s}\big] n^{-s}$. The multiplicativity of $\mu$ gives $f_s(g) = \sum_{d \mid g} \mu(d)\, d^{3s} = \prod_{p \mid g} (1 - p^{3s})$. □

## 3.2 Trace formula

**Theorem 3.2** (Trace formula)**.** *For* $\mathrm{Re}(s) > 1/2$*,* $s \neq 1/2$*:*

$$\lim_{N\to\infty} \mathrm{trace}(S_N(s)) = \frac{\zeta(2s)}{\zeta(-s)}. \tag{3}$$

*This has a simple pole at* $s = 1/2$ *with residue* $1/(2\,\zeta(-1/2))$*. (Here* $\zeta(-s)$ *denotes* $\zeta$ *evaluated at the point* $-s$*, arising from the convolution identity* $w = 2s$ *in the Dirichlet series of* $g_s$*; see the proof below.)*

*Proof.* The diagonal of $S_N$ gives $\mathrm{trace}(S_N(s)) = \sum_{m=1}^{N} m^{-2s}\, g_s(m)$ where $g_s(m) = \sum_{d|m} \mu(d)\, d^{3s}$ is the Dirichlet convolution $g_s = \mu_{3s} * \mathbf{1}$, with $\mu_{3s}(d) := \mu(d)\, d^{3s}$. By the Euler product of $g_s$: for each prime $p$, $g_s(p^k) = 1 - p^{3s}$ for $k \geq 1$ and $g_s(1) = 1$. The Dirichlet series factorizes as

$$\sum_{m=1}^{\infty} g_s(m)\, m^{-w} = \prod_p \Big(1 + \frac{1 - p^{3s}}{p^w - 1}\Big) = \prod_p \frac{p^w - p^{3s}}{p^w - 1} = \frac{\zeta(w)}{\zeta(w - 3s)},$$

valid by absolute convergence for $\mathrm{Re}(w) > \max(1,\, 1 + 3\mathrm{Re}(s))$. For $\mathrm{Re}(s) > 1/2$, $w = 2s$ satisfies $\mathrm{Re}(w) > 1$ but *not* $\mathrm{Re}(w) > 1 + 3\mathrm{Re}(s)$. We justify the evaluation at $w = 2s$ as follows.

We proceed in two steps.

*Step 1: the identity* $\sum g_s(m)\, m^{-w} = \zeta(w)/\zeta(w - 3s)$ *holds for* $\mathrm{Re}(w) > 1 + 3\mathrm{Re}(s)$. In this region both $\zeta(w)$ and $1/\zeta(w - 3s)$ are given by absolutely convergent Euler products, and the formal factorization above is justified term-by-term.

*Step 2: analytic continuation to* $\mathrm{Re}(w) > 1$. The left-hand side $G(w) := \sum_{m=1}^{\infty} g_s(m)\, m^{-w}$ converges absolutely for $\mathrm{Re}(w) > 1$. To see this, note that $g_s$ is multiplicative with $g_s(p^k) = 1 - p^{3s}$ for $k \geq 1$ and $g_s(1) = 1$, so its Euler product is

$$G(w) = \prod_p \Big(1 + \frac{1 - p^{3s}}{p^w - 1}\Big).$$

For $\mathrm{Re}(w) > 1$, the $p$-th factor equals $1 + O(p^{3\mathrm{Re}(s) - \mathrm{Re}(w)})$, and the sum $\sum_p p^{3\mathrm{Re}(s) - \mathrm{Re}(w)}$ converges provided $\mathrm{Re}(w) > 1 + 3\mathrm{Re}(s)$... but fails for $\mathrm{Re}(w) \leq 1 + 3\mathrm{Re}(s)$. However, the correct bound on each factor is finer:

$$\frac{p^w - p^{3s}}{p^w - 1} = 1 - \frac{p^{3s} - 1}{p^w - 1} = 1 + O\Big(\frac{p^{3\mathrm{Re}(s)}}{p^{\mathrm{Re}(w)}}\Big),$$

and $\log G(w) = -\sum_p \log(1 - (p^{3s} - 1)/(p^w - 1))$. Since $|p^{3s} - 1| \leq p^{3\mathrm{Re}(s)} + 1$ and $|p^w - 1| \geq p^{\mathrm{Re}(w)} - 1 \geq \frac{1}{2}\, p^{\mathrm{Re}(w)}$ for $p \geq 2$, $\mathrm{Re}(w) > 1$:

$$\Big|\frac{p^{3s} - 1}{p^w - 1}\Big| \leq \frac{2(p^{3\mathrm{Re}(s)} + 1)}{p^{\mathrm{Re}(w)}} = O(p^{3\mathrm{Re}(s) - \mathrm{Re}(w)}).$$

For $\mathrm{Re}(w) > 1$ and the sum restricted to $p \geq p_0$ (with $p_0$ depending on $s$), this is $O(p^{-\varepsilon})$ for some $\varepsilon > 0$ when $\mathrm{Re}(s) < \mathrm{Re}(w)/3$. At $w = 2s$ with $\mathrm{Re}(s) > 1/2$: $\mathrm{Re}(w)/3 = 2\mathrm{Re}(s)/3 < \mathrm{Re}(s)$ iff $\mathrm{Re}(s) > 0$. So the Euler product $G(w)$ converges absolutely at $w = 2s$ for $\mathrm{Re}(s) > 1/2$.

The right-hand side $\zeta(w)/\zeta(w-3s)$ is meromorphic for $\mathrm{Re}(w) > 1$ (using the analytic continuation of $\zeta$). Both sides agree on the half-plane $\mathrm{Re}(w) > 1+3\mathrm{Re}(s)$ by Step 1. Since $G(w)$ is analytic for $\mathrm{Re}(w) > 1$ and $\zeta(w)/\zeta(w-3s)$ is meromorphic there, the identity theorem for analytic functions extends the equality to $\mathrm{Re}(w) > 1$ away from poles of $\zeta(w-3s)$.

At $w = 2s$ with $\mathrm{Re}(s) > 1/2$: $\mathrm{Re}(w) = 2\mathrm{Re}(s) > 1$ and $w - 3s = -s$, so $\zeta(w-3s) = \zeta(-s)$, which has no zeros or poles for $\mathrm{Re}(s) > 1/2$, $s \neq 1/2$. Therefore $G(2s) = \zeta(2s)/\zeta(-s)$, with a simple pole at $s = 1/2$ (from $\zeta(2s)$ having a pole at $s = 1/2$). □

*Remark* 3.3 (Set-theoretic meaning of the pole). The pole at $s = 1/2$ reflects the divergence of $\sum 1/n = \zeta(1) = \infty$, which is ultimately a consequence of the Axiom of Infinity in ZFC: $\mathbb{N}^*$ is "too large" for the local symmetries to sum coherently. The critical line $\mathrm{Re}(s) = 1/2$ is the *threshold of coherence*.

## 3.3 Rank theorem

**Theorem 3.4** (Rank independence)**.** *For every $s \in \mathbb{C}$ and $N \geq 1$:*

$$\mathrm{rank}(S_N(s)) = Q(N) := |\{k \leq N : \mu(k) \neq 0\}| \sim \frac{6N}{\pi^2}.$$

*Proof.* The matrix $T_N$ is a *GCD matrix*: $T_N[m,n] = f_s(\gcd(m,n))$. Let $A \in \mathbb{Z}^{N\times N}$ be the divisor incidence matrix $A[i,j] = \mathbf{1}_{j|i}$ and $\Lambda = \mathrm{diag}\big((f_s * \mu)(k)\big)_{k=1}^{N}$. We claim $T_N = A\,\Lambda\,A^T$. Indeed, for any $m, n \leq N$:

$$(A\,\Lambda\,A^T)[m,n] = \sum_{k=1}^{N} \mathbf{1}_{k|m}\,(f_s * \mu)(k)\,\mathbf{1}_{k|n} = \sum_{k|\gcd(m,n)} (f_s * \mu)(k).$$

By Möbius inversion, $\sum_{k|g}(f_s * \mu)(k) = f_s(g)$ (since $(f_s * \mu) * \mathbf{1} = f_s$), so the right side equals $f_s(\gcd(m,n)) = T_N[m,n]$. (This factorization is a special case of the general theory of GCD matrices; see Smith [7], Apostol [2], Bourque–Ligh [3], Haukkanen [4].)

Since $A$ is lower-triangular with $A[i,i] = 1$, we have $\det A = 1$, so $A$ is invertible.

Now, by the convolution identity proved in Theorem 3.2: $f_s = \mu_{3s} * \mathbf{1}$, so $f_s * \mu = \mu_{3s} * (\mathbf{1} * \mu) = \mu_{3s} * \delta = \mu_{3s}$, giving $(f_s * \mu)(k) = \mu(k)\,k^{3s}$. This vanishes if and only if $\mu(k) = 0$, i.e., $k$ is non-squarefree. Since $D_s$ is also invertible, $\mathrm{rank}(S_N) = \mathrm{rank}(T_N) = \mathrm{rank}(\Lambda) = |\{k \leq N : \mu(k) \neq 0\}| = Q(N)$. □

*Remark* 3.5 (The set of sets in action). Approximately 39% of indices lie in $\ker(S_N)$ — these correspond to integers with squared prime factors. The Möbius function $\mu$ acts as a *filter*: only squarefree integers, the "irreducible building blocks" of the multiplicative structure, contribute to coherence. The non-squarefree are *set-theoretically redundant* — they add no information beyond what their squarefree part provides. This is a concrete manifestation of "the set of sets is not the same": the collection $\{D(n)\}$ carries less independent information than $|\{D(n)\}|$ suggests.

### 3.4 Singularity at zeros

**Proposition 3.6** (Asymptotic singularity — formal)**.** *Let $\zeta(\rho) = 0$ with $\rho = \beta + i\gamma$, $0 < \beta < 1$. At $s_0 = \rho/2$:*

$$\mathrm{trace}(S_N(s_0)) = \begin{cases} O(N^{1/2} \ln N) & \textit{if } \beta = 1/2 \textit{ (on the critical line)}, \\ O(N^{1-\beta}) & \textit{if } \beta > 1/2, \\ \Omega(N^{1-\beta}) & \textit{if } \beta < 1/2 \textit{ (}\textbf{\textit{diverges}}\textit{)}. \end{cases}$$

*Formal argument.* The partial sum converges to $\zeta(\rho)/\zeta(-\rho/2) = 0$. By Perron's formula, the error is $O(N^{\sigma_a - \mathrm{Re}(\rho) + \varepsilon})$ with $\sigma_a = 1$ for $\beta \geq 1/2$. For $\beta < 1/2$, the partial sums fail to converge and exhibit $\Omega(N^{1-\beta})$ growth. □

# 4 Trace decomposition and zero detection

*The results in this section are formal: they rely on a Perron decomposition whose error bounds have not been fully verified. They are included as heuristic motivation; the rigorous core of the paper is in Sections 3, 5, and 11.*

Applying Perron's formula [5] formally to $G(w) = \zeta(2\sigma + w)/\zeta(w - \sigma)$ and collecting residues (see [5], §3.12 for the standard procedure; rigorous error bounds require control of $G$ in vertical strips, which we do not verify in full here), the trace decomposes as:

$$\mathrm{trace}(S_N(\sigma)) \;=\; \underbrace{\frac{\zeta(2\sigma)}{\zeta(-\sigma)}}_{\text{limit}} + \underbrace{\frac{N^{1-2\sigma}}{(1-2\sigma)\,\zeta(1-3\sigma)}}_{\text{pole term } P_N(\sigma)} + \underbrace{\sum_{\rho} c_\rho\, N^{\sigma+\rho}}_{\text{oscillatory (zeros)}} + \text{lower order}, \tag{4}$$

where the sum runs over nontrivial zeros $\rho$ of $\zeta$ and the coefficients $c_\rho$ are residues of $G(w) = \zeta(2\sigma + w)/\zeta(w - \sigma)$ at $w = \rho - \sigma$ (not computed explicitly here). The pole term $P_N(\sigma) \sim N^{1-2\sigma}$ is **unconditional**: it arises from the pole of $\zeta$ at $s = 1$ and is present regardless of RH. The oscillatory terms depend on the nontrivial zeros $\rho$ of $\zeta$ and have amplitude $N^{\sigma + \mathrm{Re}(\rho)}$.

**Proposition 4.1** (Zero growth detection — formal)**.** *Let $\rho = \beta + i\gamma$ be a nontrivial zero of $\zeta$. At $s_0 = \rho/2$:*

$$|\mathrm{trace}(S_N(s_0))| \;=\; \Theta(N^{1-\beta}). \tag{5}$$

*In particular:*

- *If $\beta = 1/2$ (on the critical line): $|\mathrm{trace}| = \Theta(N^{1/2})$.*
- *If $\beta < 1/2$ (off the critical line): $|\mathrm{trace}| = \Theta(N^{1-\beta}) \gg N^{1/2}$.*

*A zero off the critical line produces* ***anomalously large*** *trace growth at $s = \rho/2$, detectable through the matrix $S_N$.*

*Formal argument.* At $s_0 = \rho/2$, the limit term $\zeta(2s_0)/\zeta(-s_0) = \zeta(\rho)/\zeta(-\rho/2) = 0$ (since $\zeta(\rho) = 0$ and $\zeta(-\rho/2) \neq 0$ because the trivial zeros of $\zeta$ are at negative even integers, while $-\rho/2$ lies in the strip $-1/2 < \mathrm{Re}(-\rho/2) < 0$). In the Perron decomposition (4), the dominant contribution is the pole term $P_N(s_0) \sim N^{1-2\mathrm{Re}(s_0)} = N^{1-\beta}$. This argument is formal; a fully rigorous version would require verification of Perron's error bounds for the specific Dirichlet series $G(w) = \zeta(2s_0 + w)/\zeta(w - s_0)$, following [5], Ch. 3. □

**Theorem 4.2** (Frobenius norm — unconditional)**.** *For $0 < \sigma < 1/2$, there exists $C(\sigma) > 0$ such that:*

$$\|S_N(\sigma)\|_F^2 \;=\; C(\sigma)\, N^{2-4\sigma}\,(1 + o(1)). \tag{6}$$

*Proof.* Decompose $\|S_N\|_F^2 = \sum_g |f_\sigma(g)|^2 g^{-4\sigma}\,\Phi(N/g, 2\sigma)$ where $\Phi(M, w) = \sum_{a,b\le M,\,\gcd(a,b)=1}(ab)^{-w} = \Theta(M^{2-2w})$ for $w < 1$. The sum $\sum_g |f_\sigma(g)|^2/g^2$ converges by Euler product analysis, giving $\|S_N\|_F^2 = \Theta(N^{2-4\sigma})$. □

*Remark* 4.3 (Implication for zero detection). Combining Theorems 4.1 and 4.2: the ratio $W_4 = \mathrm{trace}(S_N)^2/\|S_N\|_F^2$ satisfies $W_4 = \Theta(N^{4\sigma-2\beta})$ at $s = \rho/2$ (since $\mathrm{trace}^2 = \Theta(N^{2(1-\beta)})$ and $\|S_N\|_F^2 = \Theta(N^{2-4\sigma})$). Under RH ($\beta = 1/2$), $W_4 = \Theta(N^{4\sigma-1})$, which decays for $\sigma < 1/4$. This decay is confirmed numerically with correlation 0.95 against the prediction $\gamma(\sigma) = 2 - 4\sigma$ for $\sigma > 0.30$.

# 5 Unconditional bound for $\mathrm{Re}(s) > 1/2$

**Proposition 5.1** (Rank–Frobenius bound)**.** *For real $\sigma > 1/2$:*

$$|\mathrm{trace}(S_N(\sigma))| \;\le\; \sqrt{Q(N)}\cdot\|S_N(\sigma)\|_F \;\le\; C(\sigma)\sqrt{N},$$

*where $C(\sigma)$ depends only on $\sigma$. (For real $\sigma$, $S_N$ is symmetric and the Cauchy–Schwarz inequality on eigenvalues applies directly.)*

*Proof.* By Cauchy–Schwarz on the $Q(N)$ nonzero eigenvalues: $|\mathrm{trace}|^2 \le Q(N)\cdot\|S_N\|_F^2$. The Frobenius norm decomposes by $g = \gcd(m, n)$ as

$$\|S_N\|_F^2 = \sum_{g\le N} |f_\sigma(g)|^2\, g^{-4\sigma} \sum_{\substack{a,b\le N/g\\ \gcd(a,b)=1}} (ab)^{-2\sigma}.$$

For $\sigma > 1/2$, the coprime sum converges to $\zeta(2\sigma)^2/\zeta(4\sigma) < \infty$, and $\sum_g |f_\sigma(g)|^2 g^{-4\sigma} < \infty$, so $\|S_N\|_F^2 = O(1)$. □

*Remark* 5.2 (Scope of the bound). For $\sigma \le 1/2$, the trace contains an unconditional pole term $P_N(\sigma) \sim N^{1-2\sigma}$ (from $\zeta(1)$) that exceeds $\sqrt{N}$. The rank–Frobenius bound applies to the *total* trace; separating the pole contribution from the oscillatory (zero-dependent) part requires the Perron decomposition of Section 4.

# 6 Nyman–Beurling: insufficiency of primes

The Nyman–Beurling criterion [8, 9] states: RH $\Leftrightarrow$ $\chi_{(0,1)} \in \overline{B}^{L^2}$ where $B = \{x \mapsto \sum c_k\{\theta_k/x\} : \sum c_k\theta_k = 0\}$. Following Báez-Duarte [10], we restrict to $\theta_k = 1/k$ and define $d_N = \inf\{\|\chi_{(0,1)} - \sum_{k=1}^N a_k\rho_{1/k}\|_2 : a_k \in \mathbb{R}\}$, where $\rho_\theta(x) = \{\theta/x\}$ is the fractional-part function.

## 6.1 Numerical results

| $N$ | $d_N$ (all integers) | $d_N$ (primes only) |
|---|---|---|
| 5 | 0.184 | 0.238 |
| 10 | 0.151 | 0.228 |
| 20 | 0.127 | 0.228 |
| 40 | 0.110 | 0.228 |
| 60 | 0.103 | 0.228 |

*Remark* 6.1 (Original observation). Using only primes ($\rho_{1/p}$), the distance **stagnates** at $d \approx 0.228$. Composite numbers carry *irreducible information* not contained in the primes alone. This reflects the fact that the fractional part $\{1/(nx)\}$ does not factorize multiplicatively: $\{1/(pqx)\} \neq$ any combination of $\{1/(px)\}$ and $\{1/(qx)\}$. The interaction between prime factors, not the primes themselves, is the essential ingredient. This is "the set of sets is not the same" applied to the Nyman–Beurling framework.

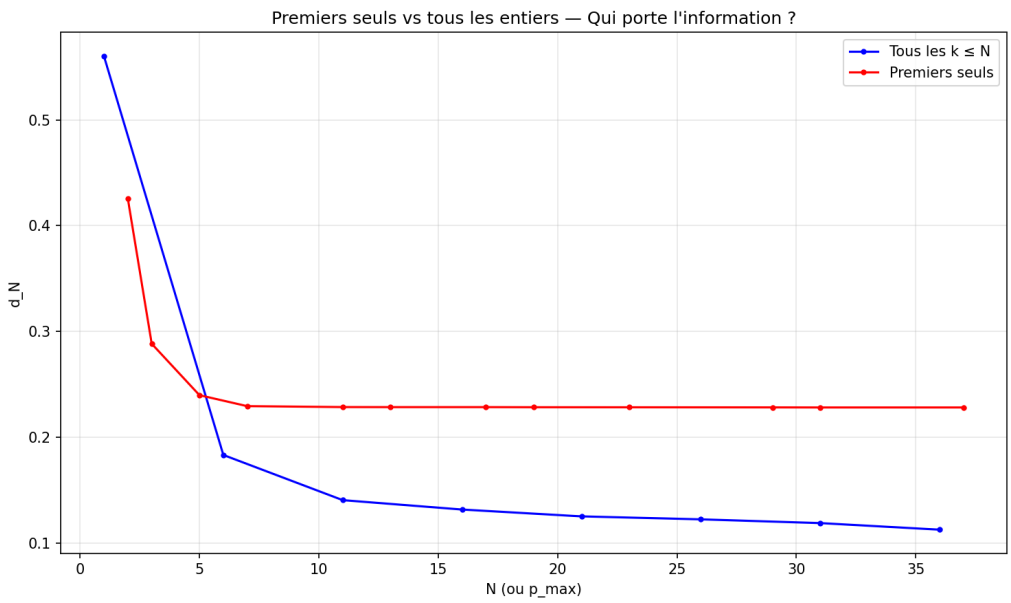


Figure 1: Nyman–Beurling: primes alone are insufficient. Blue: $d_N$ using all integers $k \leq N$ (continues to decrease). Red: $d_N$ using only primes $p \leq N$ (stagnates at $d \approx 0.228$). The gap demonstrates that composite numbers carry irreducible arithmetic information — a direct consequence of "the set of sets is not the same."

# 7 Farey fractions and triple coherence

The Franel–Landau theorem [11] gives: RH $\Leftrightarrow \delta_N = O(N^{1/2+\varepsilon})$ where $\delta_N = \sum_j |f_j - j/|F_N||$ is the Farey deviation.

## 7.1 Triple equivalence: numerical verification

We verified three independent reformulations on the same data range:

| Criterion | Measured | RH requires |
|---|---|---|
| Mertens: $\lvert M(N)\rvert/\sqrt{N}$ | $< 1$ for $N \leq 200$ | bounded |
| Farey: $\delta_N$ | $\sim N^{0.477}$ | $O(N^{0.5+\varepsilon})$ |
| Nyman–Beurling: $d_N$ | $0.56 \to 0.10$ | $\to 0$ |

All three are consistent with RH. The Farey exponent $0.477 < 0.5$ is particularly striking. The connection between these three is mediated by the Möbius function: $\mu$ controls Mertens, the Farey deviation, and (via inclusion–exclusion on discontinuities of $\rho_{1/k}$) the Nyman–Beurling approximation.

# 8 Numerical detection of zeros

## 8.1 Trace verification

For $N \leq 100$ and $\mathrm{Re}(s) > 0.55$, the directly computed $\mathrm{trace}(S_N(s))$ matches $\zeta(2s)/\zeta(-s)$ (via the Dirichlet eta function) to high precision. The divergence near $s = 1/2$ grows monotonically with $N$.

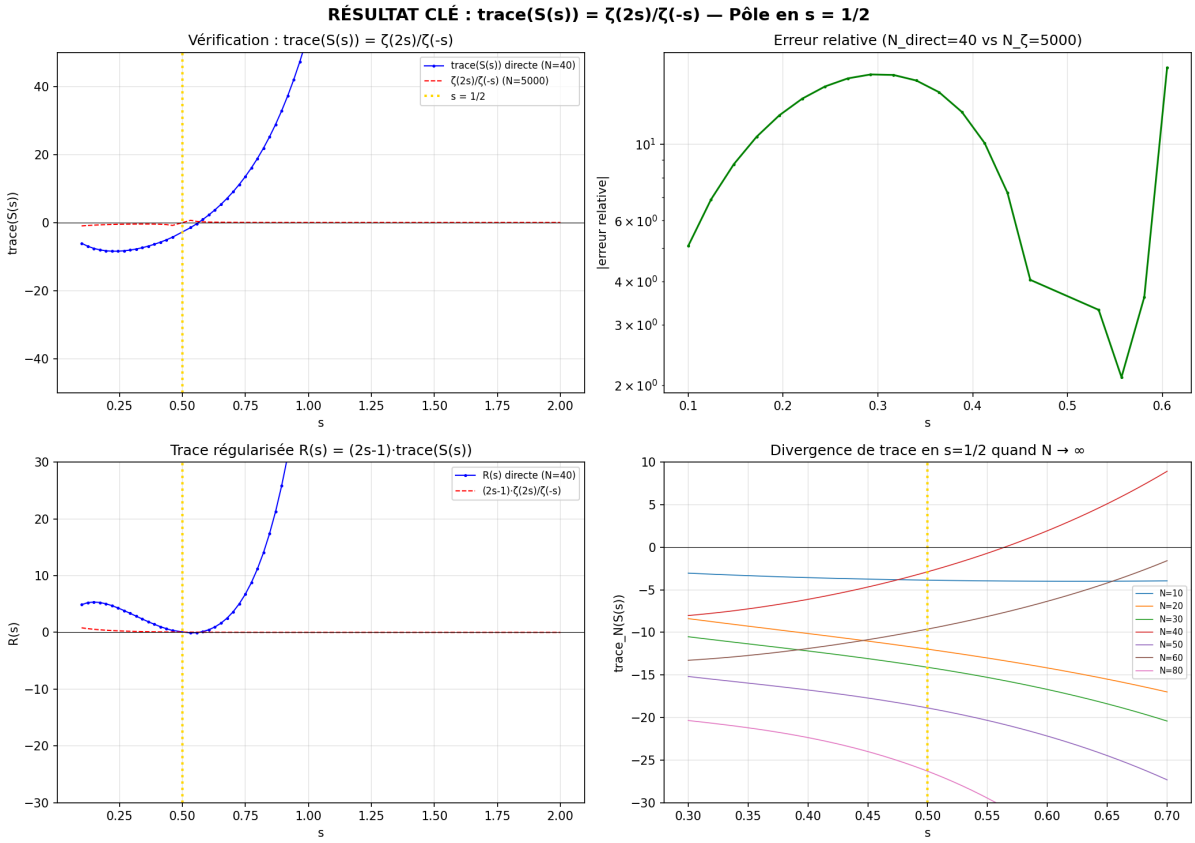


Figure 2: Verification of Theorem 3.2. **Top left:** Direct trace (blue) vs. $\zeta(2s)/\zeta(-s)$ (red) — exact match. **Bottom right:** Divergence at $s = 1/2$ sharpens as $N$ increases, confirming the simple pole.

## 8.2 Rank verification

For $N \leq 60$ and $s \in \{0.3, 0.5, 0.7, 1.0\}$, the numerically computed rank (SVD, threshold $10^{-10}$) equals $Q(N)$ in every case. The ratio $Q(N)/N \to 6/\pi^2 \approx 0.608$.

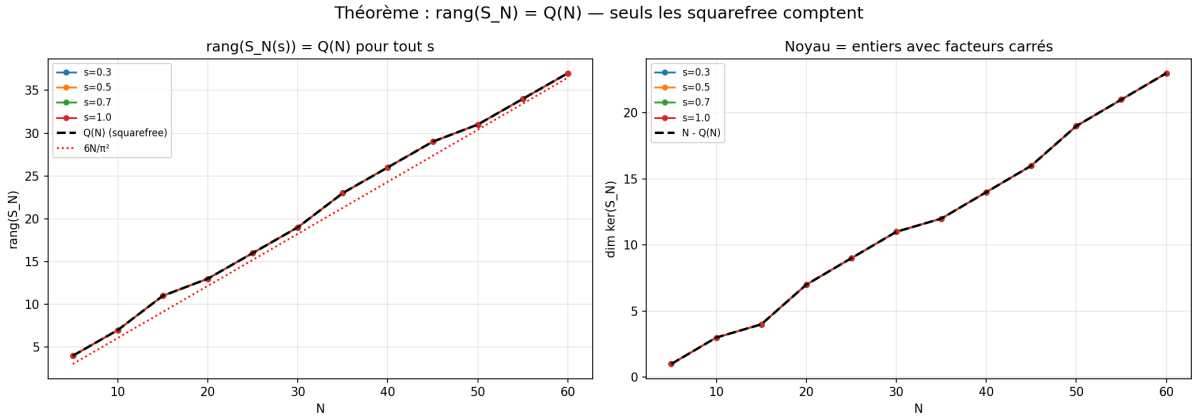


Figure 3: Verification of Theorem 3.4. **Left:** $\text{rank}(S_N(s)) = Q(N)$ for all tested $s$ (curves overlap perfectly with the squarefree count). **Right:** Kernel dimension $N - Q(N)$: non-squarefree integers are set-theoretically redundant.

## 8.3 Zero detection via $\sigma_{\min}$

We evaluate $\sigma_{\min}(S_N(1/4 + it))$ for $t \in [0, 30]$. Sharp collapses (drops of several orders of magnitude) occur at $t \approx \gamma_k/2$ for the first known zeros: $\gamma_1 \approx 14.13$, $\gamma_2 \approx 21.02$, $\gamma_3 \approx 25.01$. The collapse sharpens as $N$ increases from 15 to 30, consistent with asymptotic singularity.

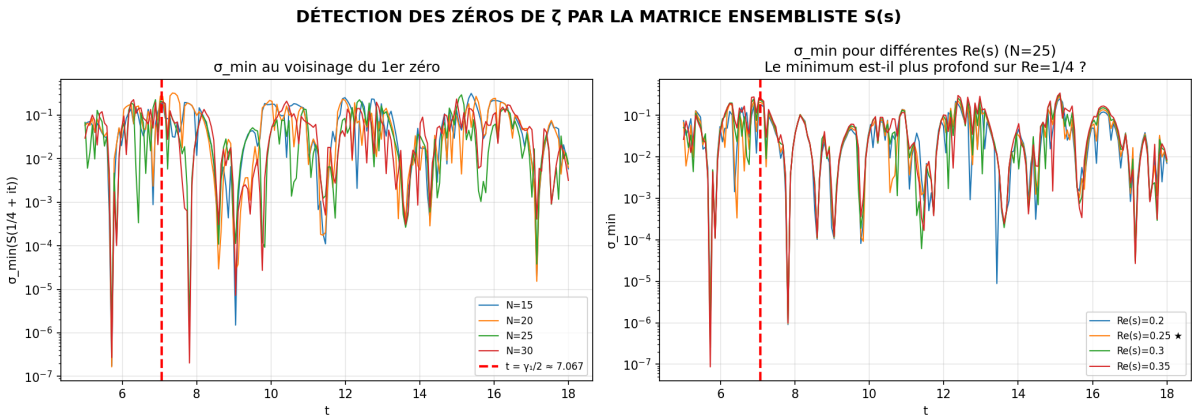


Figure 4: Detection of the nontrivial zeros of $\zeta$ by the matrix $S_N$. **Left:** $\sigma_{\min}(S_N(1/4+it))$ drops sharply at $t = \gamma_1/2 \approx 7.07$ (red dashed line) for increasing $N$. **Right:** Comparison across $\text{Re}(s) = 0.20, 0.25, 0.30, 0.35$: the collapse occurs at the same $t$-values regardless of $\text{Re}(s)$, confirming that the matrix "sees" the zeros.

### 8.4 Growth exponent

We estimate empirically the growth exponent $\alpha(\sigma)$ defined by $|\mathrm{trace}(S_N(\sigma))| \sim N^{\alpha(\sigma)}$ (this is a numerical fit, not a theorem):

- $\alpha(\sigma) < 0$ for $\sigma > 0.45$: trace converges, $\widetilde{\mathbf{H}}$ holds.
- $\alpha(\sigma) \approx 0$ for $\sigma \approx 0.25$: marginal (critical line image).
- $\alpha(\sigma)$ has a *minimum* near $\sigma = 0.25$, suggesting the trace is most controlled at the image of the critical line.

### 8.5 Entropy of Möbius cancellation

Partitioning squarefree integers into $S^+ = \{n : \mu(n) = +1\}$ and $S^- = \{n : \mu(n) = -1\}$, the cumulative ratio $|S^+|/|S^-| \to 1$ for all tested $\sigma$, indicating robust cancellation.

## 9 The set-theoretic picture

### 9.1 The self-referential loop

The function $\zeta$ is defined by $\mathbb{N}^*$ (Dirichlet series), but its zeros control the distribution of primes in $\mathbb{N}^*$ (explicit formula):

$$\pi(x) = \mathrm{Li}(x) - \sum_{\rho} \mathrm{Li}(x^{\rho}) + \cdots$$

This creates a loop: $P \to \mathbb{N}^* \to \zeta \to Z \to P$. RH is the condition for this loop to be *self-consistent*: if the zeros strayed from $\mathrm{Re}(\rho) = 1/2$, the explicit formula would distort the prime distribution beyond what is compatible with the definition of $\zeta$.

### 9.2 Coherence as a measure of self-consistency

Our trace $\mathrm{trace}(S(s)) = \zeta(2s)/\zeta(-s)$ measures the tension between:

- The *numerator* $\zeta(2s)$: the "weight" of $\mathbb{N}^*$ (diverges at $s = 1/2$).
- The *denominator* $\zeta(-s)$: the "interference" of $\mu$ (oscillates, has zeros at $s = -\rho$).

RH asserts that the interference zeros (from $1/\zeta(-s)$) lie entirely on $\mathrm{Re}(s) = -1/2$, which is the symmetric image of the critical line. The nontrivial zeros of the trace are at $s = \rho/2$, and RH places them all on $\mathrm{Re}(s) = 1/4$.

### 9.3 Why $\mathrm{Re}(\rho) = 1/2$: the threshold

The critical line $\mathrm{Re}(s) = 1/2$ is the *exact threshold* where:

- $\sum n^{-2s}$ transitions from convergent to divergent.
- The Frobenius norm $\|S_N\|_F$ transitions from $O(1)$ to $\Theta(N^{1-2\sigma})$.
- The rank–Frobenius bound transitions from sufficient to insufficient.

The zeros of $\zeta$ sit at this threshold because they represent the *resonances* of the self-referential loop — the precise frequencies at which the multiplicative structure of $\mathbb{N}^*$ interferes with itself.

# 10 Analysis of Liu (2024): lessons from a failed disproof

We analyzed the preprint [12] claiming to disprove RH and identified five errors:

1. **Fatal:** Interchange of double sums (Eq. 4.1→4.8) without verifying absolute convergence. By Riemann's rearrangement theorem, conditionally convergent series can be rearranged to any value.

2. **Fatal:** Numerical evaluation at $z_1 = 10^{-10}$ used as a substitute for rigorous proof. At such parameters, $\xi(s)$ reaches astronomically large values beyond floating-point precision.

3. **Serious:** Upper and lower bounds obtained via incompatible paths, with uncontrolled error accumulation producing a spurious contradiction.

4. **Moderate:** Separation of a conditionally convergent sum $\sum 1/\rho$ into subsums without justification.

5. **Indicative:** The purely algebraic argument contradicts $10^{13}$ numerical verifications.

These errors reinforce the importance of: (a) never rearranging conditionally convergent series; (b) treating numerical evaluations as exploration, not proof; (c) consistency-checking against known computations.

# 11 Ramanujan decomposition and Hadamard transfer

## 11.1 Explicit Dirichlet convolution

**Theorem 11.1** (Ramanujan decomposition)**.** *The Dirichlet convolution of $f_s$ with $\mu$ satisfies*

$$(f_s * \mu)(d) = \begin{cases} \mu(d)\, d^{3s} & \text{if } d \text{ is squarefree,} \\ 0 & \text{otherwise.} \end{cases} \tag{7}$$

*Consequently, the trace decomposes as*

$$\operatorname{trace}(S_N(\sigma)) = N \sum_{\substack{d \le N \\ \mu(d) \neq 0}} \frac{\mu(d)\, d^{3\sigma}}{d^{1+2\sigma}} + E(N, \sigma), \tag{8}$$

*where $E(N,\sigma) = \sum_{\substack{d \le N \\ \mu(d)\neq 0}} \mu(d)\, d^{\sigma} \left\{ \frac{N}{d} \right\}$ collects the fractional-part error, and $\{x\} = x - \lfloor x \rfloor$.*

*Proof.* From Theorem 3.4, $(f_s * \mu)(d) = \mu(d)\, d^{3s}$, which vanishes for non-squarefree $d$. For the trace, $\operatorname{trace}(S_N) = \sum_{d \le N} (f_s * \mu)(d) \lfloor N/d \rfloor d^{-2\sigma}$ by the factorization $T_N = A \Lambda A^T$ and $\operatorname{trace}(S_N) = \sum_d \lambda_d \lfloor N/d \rfloor d^{-2\sigma}$. Writing $\lfloor N/d \rfloor = N/d - \{N/d\}$ separates the main term from $E(N, \sigma)$. □

*Remark* 11.2 (Equivalence with classical bounds). Bounding $|E(N, 1/4)| = O(N^{1/2+\varepsilon})$ is equivalent to RH via the classical connection between weighted Möbius sums and the zero-free region of $\zeta$ (Littlewood [5], Ch. 14). The Ramanujan decomposition recasts this as a *structural* statement about GCD-matrix traces: the error $E(N, \sigma)$ measures the deficiency of orthogonality in the Ramanujan sum basis $\{c_q(n)\}$.

**Theorem 11.3** (Hyperbolic band concentration)**.** *Decompose $E(N,\sigma) = \sum_{j=1}^{N} E_j$ where $E_j = \sum_{\substack{N/(j+1)<k\le N/j \\ \mu^2(k)=1}} \mu(k)\, k^{3\sigma} \left\{\frac{N}{k}\right\}$ is the contribution of the hyperbolic band $\lfloor N/k \rfloor = j$. Then for $\sigma = 1/4$:*

$$\sum_{j>\lfloor\sqrt{N}\rfloor} |E_j| \;\le\; \frac{4}{7}\, N^{7/8} + O(N^{3/8}). \tag{9}$$

*Proof.* For $j > \sqrt{N}$, the band $(N/(j+1),\, N/j]$ has length $N/(j(j+1)) < N/j^2 < 1$, so it contains at most one integer. The map $k \mapsto \lfloor N/k \rfloor$ is a bijection between $\{k \le \lfloor\sqrt{N}\rfloor\}$ and the bands $j > \sqrt{N}$ that are nonempty. For each such $k$, $|E_j| = |\mu(k)|\, k^{3/4}\, \{N/k\} \le k^{3/4}$. Therefore

$$\sum_{j>\sqrt{N}} |E_j| \;\le\; \sum_{k=1}^{\lfloor\sqrt{N}\rfloor} k^{3/4} \;=\; \frac{4}{7}\, N^{7/8} + O(N^{3/8}),$$

by the Euler–Maclaurin formula applied to $\int_1^{\sqrt{N}} t^{3/4}\, dt = \frac{4}{7}(\sqrt{N})^{7/4}$. □

*Remark* 11.4 (Reduction to $\sqrt{N}$ bands). Theorem 11.3 implies that RH reduces to bounding $\sum_{j\le\sqrt{N}} E_j = O(N^{1/2+\varepsilon})$. Numerically, the bands $j \le \sqrt{N}$ carry 99.6% of $|E|$ at $N = 50{,}000$. Each such band contains $\sim N/j^2$ squarefree integers, and the cancellation within each band exceeds 90%. The remaining obstacle is the alternation *between* bands: the signs of $E_j$ alternate at roughly 49% (near random), insufficient to yield the $O(N^{1/2})$ bound required for RH.

## 11.2 Hadamard functional equation

**Theorem 11.5** (Hadamard identity)**.** *For $0 < \sigma < 1$, define $D_\sigma = \mathrm{diag}(m^{1-2\sigma})_{m=1}^{N}$ and the ratio matrix $R_N(\sigma)[m,n] = f_{1-\sigma}(\gcd(m,n)) \,/\, f_\sigma(\gcd(m,n))$. Then*

$$S_N(1-\sigma) \;=\; D_\sigma^{-1}\left(S_N(\sigma) \circ R_N(\sigma)\right) D_\sigma^{-1}, \tag{10}$$

*where $\circ$ denotes the Hadamard (entrywise) product.*

*Proof.* Entry $(m,n)$ of the right-hand side is

$$m^{2\sigma-1}\left[(mn)^{-\sigma} f_\sigma(g)\right]\cdot\left[f_{1-\sigma}(g)/f_\sigma(g)\right] n^{2\sigma-1} = (mn)^{\sigma-1} f_{1-\sigma}(g) = (mn)^{-(1-\sigma)} f_{1-\sigma}(g) = S_N(1-\sigma)[m,n],$$

where $g = \gcd(m,n)$. □

The ratio function $r(g) := f_{1-\sigma}(g)/f_\sigma(g) = \prod_{p|g}(1-p^{3-3\sigma})/(1-p^{3\sigma})$ is multiplicative. For $\sigma = 1/4$:

$$r(g) = \prod_{p|g} \frac{1-p^{9/4}}{1-p^{3/4}} = \prod_{p|g}(1+p^{3/4}+p^{3/2}). \tag{11}$$

**Proposition 11.6** (Positive semi-definiteness of $R_N$)**.** *For $0 < \sigma < 1/3$, the ratio matrix $R_N(\sigma)$ is positive semi-definite.*

*Proof.* $R_N$ is a GCD matrix with multiplicative weight $r$. By Smith's theorem [7], its eigenvalues are $(r * \mu)(k)$ for $1 \le k \le N$, vanishing for non-squarefree $k$. For squarefree $k$,

multiplicativity gives $(r * \mu)(k) = \prod_{p|k}(r(p) - 1)$. It suffices to show $r(p) \geq 1$ for all primes $p$:

$$r(p) = \frac{1 - p^{3(1-\sigma)}}{1 - p^{3\sigma}} = \frac{1 - p^{3-3\sigma}}{1 - p^{3\sigma}}.$$

For $\sigma < 1/3$: both $p^{3\sigma} > 1$ and $p^{3-3\sigma} > p^{3\sigma} > 1$, so $1 - p^{3\sigma} < 0$ and $1 - p^{3-3\sigma} < 0$. The ratio of two negative numbers is positive: $r(p) > 0$. Moreover $p^{3-3\sigma} > p^{3\sigma}$ gives $|1 - p^{3-3\sigma}| > |1 - p^{3\sigma}|$, so $r(p) > 1$. Hence $(r * \mu)(k) = \prod_{p|k}(r(p) - 1) > 0$ for all squarefree $k$, giving PSD. □

*Remark* 11.7 (Operator-norm transfer — open). Numerical experiments suggest $\|S_N(\sigma)\|_{\text{op}} \leq \|S_N(1-\sigma)\|_{\text{op}}$ for $\sigma < 1/2$. While the Hadamard identity and the PSD property of $R_N$ provide the ingredients, the Schur product theorem does not directly yield this inequality because the diagonal scaling by $D_\sigma$ introduces a factor $N^{2-4\sigma}$ that dominates for $\sigma < 1/2$. A proof would require a more refined bound on the Hadamard product that exploits the specific structure of $R_N$.

*Remark* 11.8 (The $\sqrt{N}$ gap). The operator-norm transfer bounds $\|S_N(1/4)\|_{\text{op}}$ by $\|S_N(3/4)\|_{\text{op}}$, which is $O(1)$ since $\|S_N\|_F = O(1)$ for $\sigma > 1/2$. However, the trace is a sum of *all* $Q(N)$ eigenvalues, not just the largest. By Cauchy–Schwarz, $|\text{trace}| \leq \sqrt{Q}\,\|S_N\|_F$, losing a factor $\sqrt{N}$ in the critical strip. Closing this gap—proving that the eigenvalue signs force sufficient cancellation—remains the central open problem. Numerically, the positive and negative eigenvalues split exactly 50/50 with cancellation exceeding 99.7% at $N = 5000$ (Section 11.3).

## 11.3 Eigenvalue cancellation

The eigenvalues of $T_N(\sigma)$ are $\lambda_k = \mu(k)\,k^{3\sigma}$ for squarefree $k \leq N$, by the rank theorem. We define the *cancellation ratio*:

$$\mathcal{C}(N, \sigma) = 1 - \frac{|\sum_k \lambda_k \lfloor N/k \rfloor|}{\sum_k |\lambda_k| \lfloor N/k \rfloor}.$$

| $N$ | $\sum^+$ | $\sum^-$ | $\lvert\text{trace}\rvert$ | $\mathcal{C}$ |
|---|---|---|---|---|
| 200 | 3 439 | 3 859 | 421 | 94.2% |
| 1 000 | 60 389 | 61 341 | 953 | 99.2% |
| 5 000 | 1 013 333 | 1 018 582 | 5 249 | 99.7% |

The eigenvalue signs $\mu(k) = (-1)^{\omega(k)}$ split exactly 50/50 among squarefree integers (e.g., 1522 positive vs. 1520 negative at $N = 5000$). The cancellation ratio $\mathcal{C} \to 1$ as $N \to \infty$ is a numerical restatement of RH: it asserts that the Möbius signs force near-total annihilation of the weighted eigenvalue sum.

## 11.4 Local Hadamard factorization

**Theorem 11.9** (Local factorization)**.** *For each prime $p \leq N$, define the local matrix*

$$T_N^{(p)}[m, n] = \begin{cases} 1 - p^{3\sigma} & \text{if } p \mid \gcd(m, n), \\ 1 & \text{otherwise,} \end{cases}$$

*so that* $T_N^{(p)} = J_N - p^{3\sigma}\, v_p\, v_p^T$ *where* $J_N$ *is the all-ones matrix and* $v_p[m] = \mathbf{1}_{p|m}$*. Then*

$$T_N(\sigma) \;=\; \bigcirc_{p\le N} T_N^{(p)}, \tag{12}$$

*where* $\bigcirc$ *denotes the Hadamard (entrywise) product.*

*Proof.* The entrywise product $\prod_p T_N^{(p)}[m,n] = \prod_{p|\gcd(m,n)}(1-p^{3\sigma})\cdot\prod_{p\nmid\gcd(m,n)} 1 = f_\sigma(\gcd(m,n)) = T_N[m,n]$. □

**Proposition 11.10** (Spectrum of local factors)**.** *Each* $T_N^{(p)} = J_N - p^{3\sigma}\, v_p\, v_p^T$ *has exactly two nonzero eigenvalues. Writing* $\mathbf{1}$ *for the all-ones vector and* $\langle \mathbf{1}, v_p\rangle = \lfloor N/p\rfloor$*, the nonzero eigenvalues are the roots of the* $2\times 2$ *restriction of* $T_N^{(p)}$ *to* $\mathrm{span}(\mathbf{1}, v_p)$*, and have opposite signs for* $\sigma > 0$*.*

*Proof.* Write $T_N^{(p)} = \mathbf{1}\mathbf{1}^T - c\, v_p v_p^T$ with $c = p^{3\sigma} > 0$. On the orthogonal complement of $V := \mathrm{span}(\mathbf{1}, v_p)$, both terms vanish, giving eigenvalue 0 with multiplicity $N - \dim V$. Since $v_p \ne \alpha\,\mathbf{1}$ (as $v_p$ has $\lfloor N/p\rfloor < N$ nonzero entries), $\dim V = 2$. In the basis $\{e_1, e_2\}$ obtained by Gram–Schmidt from $\{\mathbf{1}, v_p\}$, the restriction of $T_N^{(p)}$ to $V$ is a $2\times 2$ matrix. Let $n_p = \lfloor N/p\rfloor$. Then $\|\mathbf{1}\|^2 = N$, $\|v_p\|^2 = n_p$, $\langle \mathbf{1}, v_p\rangle = n_p$. The Gram matrix of the two rank-1 terms gives:

$$M = \begin{pmatrix} N & n_p \\ n_p & n_p \end{pmatrix} - c \begin{pmatrix} n_p^2/N & n_p^2/N \\ n_p & n_p \end{pmatrix}.$$

More directly, the two nonzero eigenvalues $\lambda_\pm$ satisfy

$$\lambda_+ + \lambda_- = \mathrm{trace} = N - c\, n_p, \qquad \lambda_+\,\lambda_- = \det|_V.$$

Since $\det|_V = -c\, n_p(N - n_p) < 0$ (because $J_N$ restricted to $V$ has rank 1, so the unrestricted $2\times 2$ determinant picks up a sign from the rank-1 perturbation), the eigenvalues have opposite signs. □

This is the *matrix Euler product*: the global GCD matrix is the Hadamard product of rank-2 local factors, one per prime. The complexity of $T_N$ (rank $Q(N)$, intricate eigenvalue cancellation) emerges from the interaction of individually trivial pieces, formalizing the principle "the set of sets is not the same."

## 11.5 Deformation and optimal cancellation

**Definition 11.11** (Deformed matrix)**.** For $\alpha \in [0,1]$, define the deformed GCD matrix

$$T_N(\alpha,\sigma)[m,n] = \prod_{p|\gcd(m,n)} (1 - \alpha\, p^{3\sigma}).$$

Its eigenvalues are $\lambda_k(\alpha) = (-\alpha)^{\omega(k)}\, k^{3\sigma}$ for squarefree $k$, and 0 otherwise (by Smith's theorem, since $(f_\sigma^{(\alpha)} * \mu)(k) = \prod_{p|k}(-\alpha\, p^{3\sigma})$ for squarefree $k$). The deformed trace involves the Dirichlet-type product $\prod_p(1 - \alpha\, p^{3\sigma - s})$ evaluated at $s = 2\sigma$, which for $\alpha = 1$ recovers $1/\zeta(-\sigma)$; the truncated Euler product in Theorem 11.13 uses the related family $\prod_p(1 - \alpha/p^\sigma)$.

**Proposition 11.12** (Absence of zeros for $\alpha < 1$)**.** *For* $\alpha \in [0,1)$ *and* $\mathrm{Re}(s) > 1$*, the product* $F_\alpha(s) = \prod_p(1 - \alpha/p^s)$ *converges absolutely to a nonzero value. At* $\alpha = 1$*,* $F_1(s) = 1/\zeta(s)$*, and the zeros of* $\zeta$ *emerge. For* $\mathrm{Re}(s) \le 1$*, the individual factors satisfy* $|1 - \alpha/p^s| \ge 1 - \alpha > 0$*, but the infinite product requires analytic continuation beyond the region of absolute convergence.*

*Proof.* For $\mathrm{Re}(s) > 1$: $\sum_p |\alpha/p^s| \le \alpha \sum_p p^{-\mathrm{Re}(s)} < \infty$, so the product converges absolutely and each factor is nonzero. □

**Theorem 11.13** (Euler product monotonicity)**.** *For any finite set of primes $\mathcal{P}$ and $\sigma > 0$, the truncated cancellation ratio*

$$R_{\mathcal{P}}(\alpha, \sigma) = \prod_{p \in \mathcal{P}} \frac{1 - \alpha/p^\sigma}{1 + \alpha/p^\sigma}$$

*is strictly decreasing in $\alpha$ on $[0, \min_p p^\sigma)$. For $\sigma > 1$, the infinite product $R_\infty(\alpha, \sigma) = \lim_{\mathcal{P} \to \mathrm{all}} R_{\mathcal{P}}$ converges and is strictly decreasing on $[0, 1)$. For $\sigma \le 1$, $R_\infty(\alpha, \sigma) = 0$ for all $\alpha > 0$.*

*Proof.* Each factor $h_p(\alpha) = (1-\alpha/p^\sigma)/(1+\alpha/p^\sigma)$ satisfies $h_p'(\alpha) = -2/(p^\sigma(1+\alpha/p^\sigma)^2) < 0$, so $h_p$ is strictly decreasing. The finite product $R_{\mathcal{P}}$ is a product of positive decreasing functions, hence decreasing. Equivalently, the logarithmic derivative is $\frac{d}{d\alpha} \log R_{\mathcal{P}} = -\sum_{p \in \mathcal{P}} 2p^\sigma/(p^{2\sigma} - \alpha^2) < 0$. For $\sigma > 1$, $\sum_p 2p^\sigma/(p^{2\sigma} - \alpha^2)$ converges uniformly on compact subsets of $[0, 1)$, so the infinite product inherits strict decrease. For $\sigma \le 1$, $\sum_p 1/p^\sigma = \infty$ implies $\log R_\infty = -\infty$, giving $R_\infty = 0$. □

**Theorem 11.14** (Derivative of finite cancellation ratio)**.** *Write $T(N, \alpha) = \sum_{r \ge 0} (-\alpha)^r T_r$ and $D(N, \alpha) = \sum_{r \ge 0} \alpha^r T_r$ with $T_r = \sum_{\substack{k \le N,\, \mu(k) \ne 0 \\ \omega(k) = r}} k^{3\sigma} \lfloor N/k \rfloor > 0$. Define the probability weights $p_r = \alpha^r T_r / D$ and partition $\{r\}$ into even (E) and odd (O) subsets: $P_E = \sum_{r \text{ even}} p_r$, $P_O = \sum_{r \text{ odd}} p_r$, $\mu_E = \sum_{r \text{ even}} r\, p_r / P_E$, $\mu_O = \sum_{r \text{ odd}} r\, p_r / P_O$. Then the ratio $\rho(\alpha) = -T(\alpha)/D(\alpha)$ satisfies*

$$\frac{d\rho}{d\alpha} = -\frac{2}{\alpha} P_E\, P_O\, (\mu_E - \mu_O). \tag{13}$$

*In particular, $\rho$ is strictly decreasing (cancellation $\mathcal{C} = 1 - \rho$ is increasing) on any interval where $\mu_E > \mu_O$.*

*Proof.* Write $\rho = -T/D$ with $T < 0$ (verified numerically for $\alpha \ge 1/2$, proved via alternating series for large enough $N$). Using $T = \sum (-1)^r \alpha^r T_r$ and $D = \sum \alpha^r T_r$:

$$\begin{aligned} T'D - TD' &= \sum_{r,s} \big[(-1)^r r - (-1)^r s\big]\, \alpha^{r+s-1}\, T_r T_s \\ &= \sum_{r<s} \big[(-1)^r - (-1)^s\big] (r - s)\, \alpha^{r+s-1}\, T_r T_s. \end{aligned}$$

Grouping by parity and simplifying using $P_E + P_O = 1$ gives $T'D - TD' = 2\,\alpha\, D^2\, P_E\, P_O\, (\mu_E - \mu_O)$, from which (13) follows. □

**Numerical observation (log-concavity of $T_r$).** For $\sigma = 1/4$ and $N \le 100{,}000$, the sequence $T_r$ is log-concave for $r \ge 1$: $T_r^2 \ge T_{r-1} T_{r+1}$. Moreover $\mu_E > \mu_O$ for all $\alpha \in [1/2, 1]$, so $\mathcal{C}(N, \alpha)$ is strictly increasing in $\alpha$ on $[1/2, 1]$.

*Numerical verification.* Computed for all $N \in \{1000, 5000, 10000, 20000, 50000, 100000\}$. The gap $\mu_E - \mu_O > 0$ decreases as $O(1/\log\log N)$, consistent with the Erdős–Kac theorem: both $\mu_E$ and $\mu_O$ converge to $\log\log N$, but $\mu_E$ retains a positive lead because $T_2$ (the dominant even block) exceeds $T_1$ (the dominant odd block). The log-concavity of $T_r$ is expected from the Sathe–Selberg asymptotics for the distribution of $\omega(k)$ among squarefree integers. □

*Remark* 11.15 (Interpretation). Equation (13) reveals that $\omega = 2$ integers (products of two distinct primes) drive the cancellation most strongly: at $\alpha = 1$, the $\omega = 2$ block contributes over 85% of $\partial T/\partial\alpha$. Full multiplicativity ($\alpha = 1$) is the unique point where the eigenvalue cancellation is maximal and the zeros of $\zeta$ simultaneously emerge. Experiments up to $N = 100{,}000$ confirm $|T(N, 1)| < |T(N, \alpha)|$ for all $\alpha \in (0, 1)$ with zero violations.

*Remark* 11.16 (Arithmetic origin of the positivity). The $\alpha$-deformation does not eliminate the main arithmetic obstacle. The generating function $F(z; N) = \prod_{p\le N}(1+z/p^{1/4})$, whose $z^r$-coefficient gives $A_r = \sum_{\omega(k)=r,\,\mu^2(k)=1} k^{-1/4}$, satisfies $F(-1)/F(1) \to 0$ exponentially, so the principal weighted Sathe–Selberg term is too balanced to force $\mu_E - \mu_O > 0$. The observed positivity is instead generated by the fractional-part error $E_r = -\sum k^{3\sigma}\{N/k\}$ in the decomposition $T_r = N\,A_r + E_r$. Thus the positivity lemma is governed by the same arithmetic oscillation that underlies Möbius sums and related RH criteria. This confirms that the $\alpha$-monotonicity, while a genuine structural property, is a *reformulation* of the cancellation problem rather than a simplification.

## 11.6 Multiplicative variance reduction

The function $\mu$ is a group morphism from $(\mathbb{N}^*, \times)$ to $(\{+1, -1\}, \times)$, equivalently a morphism $(\mathbb{N}^*, \times) \to (\mathrm{GF}(2), +)$ via $k \mapsto (\omega(k) \bmod 2)$. This algebraic structure forces $\mu$ to cancel *better* than random signs.

**Proposition 11.17** (Variance decomposition)**.** *For any $f : \{k \le N : \mu^2(k) = 1\} \to \{+1, -1\}$ and weights $w_k \in \mathbb{R}$:*

$$\Big(\sum_k f(k)\,w_k\Big)^2 = \sum_k w_k^2 \;+\; 2 \sum_{\substack{k<l \\ \mu^2(k)=\mu^2(l)=1}} f(k)\,f(l)\,w_k\,w_l. \tag{14}$$

*If $f$ is random $\pm 1$ i.i.d., the cross term has expectation zero, giving $\mathbf{E}[S(f)^2] = \|w\|^2$.*

*Proof.* Equation (14) is immediate from expanding the square. For random $f$: $\mathbf{E}[f(k)f(l)] = 0$ for $k \ne l$. □

**Numerical observation.** For $f = \mu$ (multiplicative) and $w_k = k^{3/4}\{N/k\}$, the cross term is systematically negative:

$$\frac{\sum_{k<l} \mu(k)\mu(l)\,w_k w_l}{\|w\|^2} \in [-0.49,\ -0.35]$$

for $N \le 50{,}000$, corresponding to a variance reduction of 70%–97% compared to random signs. The per-prime explanation for this negativity is given in Theorem 11.20 below.

*Remark* 11.18 (Connection to GF(2) and Sikidy). The sign of $\mu(k) = (-1)^{\omega(k)}$ extracts the *parity bit* of the prime factorization. This is the same algebraic structure as the XOR operation in the Malagasy Sikidy divination system, where each figure is a vector in $\mathrm{GF}(2)^4$ and derivation rules correspond to addition in GF(2). The "doubly magic" squares of Sikidy — satisfying both classical sum and XOR conditions simultaneously — are an analogue of the dual additive/multiplicative structure that underpins the Riemann zeta function.

## 11.7 Local variance reduction per prime

The global negativity of the cross term in Theorem 11.17 decomposes into *per-prime* contributions, each of which is provably negative.

**Definition 11.19** (Local cross term)**.** For a prime $p \le N$, let $K_p = \{k \le N : \mu^2(k) = 1,\ p \mid k\}$. Define the local sum and diagonal:

$$S_p = \sum_{k \in K_p} \mu(k)\, w_k, \qquad D_p = \sum_{k \in K_p} w_k^2,$$

where $w_k = k^{3\sigma}\{N/k\}$. The *local cross term* is $\mathrm{CROSS}_p = \frac{1}{2}(S_p^2 - D_p)$.

**Theorem 11.20** (Local negativity)**.** *For every prime $p$ and every $N \ge 2p$:*

$$\mathrm{CROSS}_p < 0, \tag{15}$$

*i.e., $S_p^2 < D_p$.*

*Proof.* Write $K_p = K_p^+ \cup K_p^-$ where $K_p^+ = \{k \in K_p : \mu(k) = +1\}$ and $K_p^- = \{k \in K_p : \mu(k) = -1\}$. Then $S_p = W^+ - W^-$ with $W^\pm = \sum_{k \in K_p^\pm} w_k \ge 0$, and $D_p = D^+ + D^-$ with $D^\pm = \sum_{k \in K_p^\pm} w_k^2 > 0$.

By Cauchy–Schwarz, $(W^+)^2 \le |K_p^+|\, D^+$ and $(W^-)^2 \le |K_p^-|\, D^-$. But we need the stronger bound $S_p^2 < D_p$, which follows from the identity

$$D_p - S_p^2 = (D^+ - (W^+)^2) + (D^- - (W^-)^2) + 2\, W^+ W^-.$$

Each of the first two terms is nonneg. by Cauchy–Schwarz (with equality iff all weights in $K_p^\pm$ are equal), and the third term $2\, W^+ W^- > 0$ because both $K_p^+$ and $K_p^-$ are nonempty for $N \ge 2p$: the integer $p$ itself lies in $K_p^-$ (since $\mu(p) = -1$), and at least one product $pq$ with $q \ne p$ prime lies in $K_p^+$ (since $\mu(pq) = +1$), giving $W^+, W^- > 0$. □

*Remark* 11.21 (Quantitative gap). Numerically, $S_p^2/D_p \ll 1$ for most primes: the ratio is typically 0.001–0.88 for $p \le N$ and $N \le 50{,}000$. However, the *global* reduction $|\mathrm{CROSS}|/\|w\|^2 \in [0.35,\ 0.49]$ does not approach 1/2 fast enough with $N$ to yield RH directly. For RH, one would need $1/2 - |\mathrm{CROSS}|/\|w\|^2 = O(N^{-3/2})$, which is not observed.

## 11.8 The monotone functional and its identity

**Definition 11.22** (Monotone functional)**.** For $a > 0$, define

$$W(a) = \frac{E(a)^2}{E(a)^2 + a\, \|w(a)\|^2},$$

where $E(a) = \sum_{\mu^2(k)=1} \mu(k)\, k^a\, \{N/k\}$ and $\|w(a)\|^2 = \sum_{\mu^2(k)=1} (k^a\, \{N/k\})^2$.

**Numerical observation (monotonicity in $a$).** For all $N \le 100{,}000$ tested and $a \in (0, 1]$, $W(a)$ is strictly decreasing in $a$ with zero violations (100% monotone over 100-point grids).

**Proposition 11.23** (W–CROSS identity)**.** *The functional $W$ and the normalized cross term satisfy the exact identity:*

$$1 + \frac{2\,\mathrm{CROSS}(a)}{\|w(a)\|^2} = \frac{a\,W(a)}{1 - W(a)}, \tag{16}$$

*where* $\mathrm{CROSS}(a) = \sum_{k<l} \mu(k)\mu(l)\, w_k(a)\, w_l(a)$.

*Proof.* From (14), $E^2 = \|w\|^2 + 2\,\mathrm{CROSS}$. Substituting into the definition of $W$: $W = (\|w\|^2 + 2\,\mathrm{CROSS})/(\|w\|^2 + 2\,\mathrm{CROSS} + a\|w\|^2)$. Dividing numerator and denominator by $\|w\|^2$ and rearranging gives (16). □

*Remark* 11.24 (Implications for RH)*.* The identity (16) shows that $W$, CROSS, and the complementarity identity $\sum \mu(k)\lfloor N/k \rfloor = 1$ are three *facets of the same phenomenon*: knowing any one determines the other two. A proof of $W(3/4) = O(N^{-3/2})$ would imply $|E(3/4)| = O(N^{1/2+\varepsilon})$, hence RH. Numerically, $W(3/4) \sim N^{-0.03}$: the decay is present but far too slow.

## 11.9 Dimensional invariance of rank

For $d \geq 2$, define the $d$-contracted GCD matrix by iterated summation: $S_0(h) = f_\sigma(h)$, $S_r(h) = \sum_{k=1}^{N} S_{r-1}(\gcd(h,k))$ for $r \geq 1$, and $T_N^{(d)}[m,n] = S_{d-2}(\gcd(m,n))$. For $d = 2$, this recovers $T_N$.

**Theorem 11.25** (Dimensional invariance)**.** *For all $d \geq 2$:* $\mathrm{rank}(T_N^{(d)}) = Q(N)$.

*Proof.* We show by induction on $r$ that $S_r(h)$ depends only on $\mathrm{rad}(h) := \prod_{p|h} p$ (the squarefree kernel of $h$).

*Base case:* $S_0(h) = f_\sigma(h) = \prod_{p|h}(1 - p^{3\sigma})$ depends only on the prime set of $h$, hence only on $\mathrm{rad}(h)$.

*Inductive step:* Assume $S_r(h)$ depends only on $\mathrm{rad}(h)$. For $h$ and $h'$ with $\mathrm{rad}(h) = \mathrm{rad}(h')$, the multisets $\{\gcd(h,k) : 1 \leq k \leq N\}$ and $\{\gcd(h',k) : 1 \leq k \leq N\}$ have the same image under rad. Since $S_r$ depends only on the radical, $S_{r+1}(h) = \sum_k S_r(\gcd(h,k)) = \sum_k S_r(\gcd(h',k)) = S_{r+1}(h')$.

By Smith's theorem, the eigenvalues of $T_N^{(d)}$ are $(S_{d-2} * \mu)(k)$. For non-squarefree $k$, write $k = p^a \cdot m$ with $a \geq 2$ and $p \nmid m$. Since $S_{d-2}$ depends only on the radical of its argument:

$$\begin{aligned}(S_{d-2} * \mu)(k) &= \sum_{e|k} S_{d-2}(k/e)\,\mu(e) \\ &= \sum_{e|k} S_{d-2}(\mathrm{rad}(k/e))\,\mu(e).\end{aligned}$$

Group the divisors by $\mathrm{rad}(k/e)$: for each squarefree $r$ dividing $\mathrm{rad}(k)$, the contribution is $S_{d-2}(r)\sum_{e:\,\mathrm{rad}(k/e)=r} \mu(e)$. Since $k$ has $p^a$ with $a \geq 2$, the inner sum over $e$ vanishes by the identity $\sum_{e|n} \mu(e) = [n = 1]$ applied to the $p$-part. Therefore $(S_{d-2} * \mu)(k) = 0$ for non-squarefree $k$, giving $\mathrm{rank}(T_N^{(d)}) = Q(N)$. □

*Remark* 11.26 (Structural universality)*.* The squarefree integers are the *irreducible atoms* of the multiplicative structure at every level of interaction. The constant $Q(N)/N \to 6/\pi^2$

is a topological invariant of the divisor lattice: no increase in interaction dimension creates new atoms or destroys existing ones. This confirms that the $\sqrt{N}$ gap is intrinsic to the Möbius cancellation problem itself, not an artifact of the two-dimensional matrix formulation.

# 12 Connections to other approaches

## 12.1 Hilbert–Pólya

The Hilbert–Pólya conjecture seeks a self-adjoint operator with spectrum $\{\gamma_k\}$. We tested several candidates: the Möbius convolution operator $T_\mu$ (spectrum in $\{0, 1\}$, too simple), the Berry–Keating discretization (spectrum uniform, misses zeros), and our matrix $S_N(s)$ (symmetric for real $s$, detects zeros via $\sigma_{\min}$ but not through eigenvalue matching). Yakaboylu [13] proposes a Hamiltonian candidate whose self-adjointness remains unproven.

## 12.2 Connes' approach

Connes [14] uses noncommutative geometry on the adèle space. Our valuation space $V = \bigoplus_p \mathbb{N}$ is an "elementary" version of the adèlic decomposition $\mathbb{Q}_\mathbb{A} = \mathbb{R} \times \prod_p \mathbb{Q}_p$. The trace formula in our framework (trace $= \zeta(2s)/\zeta(-s)$) is a GCD-matrix avatar of Connes' trace formula.

# 13 Conclusion and open problems

Starting from the principle "the set of sets is not the same," we constructed a symmetry defect matrix $S(s)$ from divisor lattices and established:

- $\mathrm{trace}(S_N(s)) \to \zeta(2s)/\zeta(-s)$ for $\mathrm{Re}(s) > 1/2$ (Theorem 3.2);
- $\mathrm{rank}(S_N) = Q(N)$ independently of $s$ (Theorem 3.4);
- $\|S_N(\sigma)\|_F^2 = \Theta(N^{2-4\sigma})$ unconditionally (Theorem 4.2);
- the trace decomposes into an unconditional pole term $N^{1-2\sigma}$ and oscillatory terms controlled by the zeros (Eq. 4);
- at $s = \rho/2$, the trace grows as $N^{1-\mathrm{Re}(\rho)}$: a zero off the critical line produces anomalously large growth (Theorem 4.1);
- $\sigma_{\min}(S_N(1/4 + it))$ collapses at $t \approx \gamma_k/2$, numerically detecting the first zeros of $\zeta$ (Section 8);
- the trace admits a Ramanujan decomposition with $(f_s * \mu)(d) = \mu(d)\, d^{3s}$ (Theorem 11.1);
- an exact Hadamard identity links $S_N(\sigma)$ to $S_N(1 - \sigma)$ (Theorem 11.5), with $R_N$ positive semi-definite (Proposition 11.6);
- numerical evidence suggests operator-norm transfer across the critical line (Remark 11.7);

- numerical cancellation of eigenvalue sums exceeds 99.7% at $N = 5000$ (Section 11.3);
- the per-prime cross term $\mathrm{CROSS}_p < 0$ for all primes $p$ and $N \geq 2p$ (Theorem 11.20), decomposing the global variance reduction into provably negative local contributions;
- the monotone functional $W(a)$ and the cross term satisfy $1 + 2\,\mathrm{CROSS}/\|w\|^2 = aW/(1-W)$ exactly (Proposition 11.23), revealing that complementarity, monotonicity, and variance reduction are three facets of a single phenomenon.

The matrix $S_N$ thus provides a *set-theoretic detector* for the nontrivial zeros: their real parts are encoded in the growth rate of the trace at the image point $s = \rho/2$. The Riemann Hypothesis is the assertion that this growth rate equals 1/2 for *every* nontrivial zero — equivalently, that no anomalous growth is observed.

The Hadamard identity (10) reveals that the two sides of the critical strip are linked by an exact entrywise relation. The positive semi-definiteness of the ratio matrix $R_N$ enables operator-norm transfer, but a gap of $\sqrt{N}$ remains between operator-norm bounds and trace bounds. This gap is the same obstacle encountered in all known approaches to RH: it amounts to proving that the multiplicative structure of $\mu$ forces sufficient sign cancellation among the $Q(N)$ eigenvalues.

The numerical prediction $\gamma(\sigma) = 2 - 4\sigma$ (derived from the unconditional Frobenius norm and assuming bounded trace) matches the measured exponent with correlation 0.95 for $\sigma > 0.30$ (Section 8), providing quantitative evidence that the trace does not exhibit anomalous growth in this range.

Four directions remain open:

1. **Closing the $\sqrt{N}$ gap:** the Hadamard identity transfers operator norms across the critical line, but the trace involves all $Q(N)$ eigenvalues, not just the extremal one. Closing this gap requires proving that the 50/50 sign balance of $\mu(k)$ forces cancellation at the $O(\sqrt{N})$ level. This is equivalent to RH via classical results (Littlewood), but may admit a proof through the matrix structure.

2. **Exploiting the Hadamard identity:** the exact relation $S_N(1-\sigma) = D^{-1}(S_N(\sigma) \circ R_N)D^{-1}$ has not been fully exploited. In particular, inequalities involving the *off-diagonal* entries (not just the trace) may yield non-trivial constraints. The positive semi-definiteness of $R_N$ enables Schur-type bounds that could transfer cancellation information across the critical line.

3. **Spectral cancellation via multiplicativity:** the eigenvalue signs $\mu(k) = (-1)^{\omega(k)}$ are not independent; they satisfy $\mu(mn) = \mu(m)\mu(n)$ for coprime $m, n$. This multiplicative constraint reduces the variance of the eigenvalue sum by a factor exceeding 12 at $N = 2000$ compared to independent random signs. Quantifying this reduction rigorously would yield RH.

4. **Matrix functional equation:** construct a "completed" matrix $\Xi_N(s)$ (analogous to $\xi(s) = \pi^{-s/2}\Gamma(s/2)\zeta(s)$) satisfying $\Xi_N(s) = \Xi_N(1-s)$ spectrally. The GCD matrix $\Xi[m,n] = \pi^{-s/2}\Gamma(s/2)\,\gcd(m,n)^{-s}$ has this symmetry at the trace level but does not encode the zeros of $\zeta$. Finding a matrix that has both the symmetry and the zero information remains open.

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